

YUSUF AHMED KHAN

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SUMMARY

PhD student (Penn State EE) working on efficient and reliable machine learning: training algorithms that skip the memory-heavy backward pass, software-level repair for AI models corrupted by hardware faults, and retrieval-augmented generation. Hands-on experience building RAG pipelines (Sentence Transformers, FAISS, HuggingFace) and implementing LLM architectures; coursework in large language models.

EDUCATION

PhD in Electrical Engineering

Aug 2024 – May 2029 (expected)

The Pennsylvania State University

GPA: 4.0 / 4.0

Adviser: [Dr. Abhronil Sengupta](#), Neuromorphic Computing Lab

Relevant Coursework: Probability & Stochastic Processes; Introduction to Neural Networks; Socially Responsible AI: Large Language Models

SKILLS

Languages Python, MATLAB

Frameworks PyTorch, NumPy, SciPy, Pandas, Matplotlib, HuggingFace Transformers, Sentence Transformers, FAISS, scikit-learn

Systems CUDA, Linux CLI, Git, VSCode, MobaXterm

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2024 – Present

The Pennsylvania State University | Adviser: [Dr. Abhronil Sengupta](#)

State College, PA

- Developed **STAR** ([arXiv:2607.15415](#)): a retraining-based repair method for AI models deployed on memory chips that develop permanent hardware faults, requiring no additional hardware or fault localization. Before deployment, a compact snapshot of the healthy model's internal states is stored (**20×–191× smaller** than a full checkpoint); after faults appear, this snapshot guides retraining to recover accuracy without any knowledge of which memory cells are broken. Proved the augmented training dynamics retain fixed-point convergence guarantees under Lipschitz-continuity conditions. Achieves **+59%** accuracy recovery over standard retraining on CNN and MLP benchmarks.
- Working on scaling Equilibrium Propagation (EP), a local learning rule that computes weight updates without a backward pass, to deeper network architectures and developing methods to improve its computational efficiency; also investigating its theoretical foundations, including gradient-equivalence to backpropagation in appropriate limits, with a neuroscience-inspired angle drawing on glial-cell-mediated regulation of synaptic plasticity.
- Built **PyTorch/CUDA** training pipelines for fault-injection experiments: simulating permanent memory failures at rates up to 90%, convergence diagnostics, and sweeps across MLP and CNN architectures.

Research Intern

May 2022 – Dec 2022

National Remote Sensing Centre, ISRO | Supervisor: [Ms. Khushboo Mirza](#)

New Delhi, India

- Built an end-to-end ML pipeline (**Python, scikit-learn**) for industrial site detection in multispectral Sentinel-2 satellite imagery, achieving **96.4% classification accuracy** via discriminative multi-band spectral feature engineering.

SELECTED PROJECTS

STAR: Keeping AI Models Reliable When Hardware Memory Fails

[arXiv:2607.15415](#) | Jul 2025

Graduate Research Project, Penn State

- Designed an astrocyte-inspired repair mechanism for Equilibrium Propagation-trained networks under permanent stuck-at hardware faults; recovered **+59%** accuracy on CNN/CIFAR-10 and **+52%** on MLP/MNIST over standard retraining across fault rates up to **90%**, using repair signals **20×–191× smaller** than a full model checkpoint, with no fault localization required.

Neuromorphic RAG System

[GitHub](#) | *Personal Project*

- Built a retrieval-augmented generation pipeline over a corpus of ML research papers, combining **Sentence Transformers** for dense embeddings, **FAISS** for semantic search, and **Flan-T5** for grounded answer generation; implemented full ingestion pipeline from PDF extraction through FAISS index construction.

SELECTED PUBLICATIONS

- Y. A. Khan**, Z. Han, A. Sengupta, “STAR: Astrocyte-Inspired SState-Augmented Repair for Supervised Memristive AI Hardware Systems,” *arXiv:2607.15415*, Jul 2025. [\[Link\]](#)
- Y. A. Khan**, A. Mahboob, O. Farooq, “Enhancing Sleep-Wake Classification Accuracy: Unveiling the Significance of KL Divergence in EEG Signal Analysis,” *ICCCI 2024*. [\[Link\]](#)
- Y. A. Khan**, S. Imaduddin, et al., “Artificial Intelligence Based Approach for Classification of Human Activities Using MEMS Sensors Data,” *Sensors*, 23:1275, 2023. [\[Link\]](#)

HONORS & AWARDS

DAAD WISE Scholar, fully funded research fellowship, University of Kiel, Germany (competitive, international selection).

Bloom Memorial Fellowship & Waynick Graduate Scholarship in EE, Penn State (\$8,000 total).

IEEE Merwin Student Scholarship (\$1,000).